*Precisa calcular assim?

1) a) $Im = \{y \in \mathbb{R} \mid y = 2 \text{ ou } x = -2\}$ $D = \{x \in \mathbb{R} \mid x > 2 \text{ e } x < -2\}$

b) $D = \{x \in \mathbb{R} \mid x \leq 5 \text{ e } x \geq -5\}$ $Im = \{y \in \mathbb{R} \mid y \geq -4 \text{ e } x \leq 4\}$

c) $D = x \in \mathbb{R}\ [-5,-1) \cup (-1,1) \cup (1,5]$ $Im = y \in \mathbb{R}\ (1,6]$

d) $D = x \in \mathbb{R}\ [-5,5]$ $Im = y \in \mathbb{R}\ [-5,5]$

e) $D = x \in \mathbb{R}\ [-5,5]$ $Im = y \in \mathbb{R}\ [2,0] \cup (4,9]$

$D = \{x \in \mathbb{R} \mid x > -\infty \text{ e } x < \infty\}$

f) $Im = \{y \in \mathbb{R} \mid y < 1\}$

3) a) Crescente, b) Decrescente, c) Crescente, d) Decrescente, e) Crescente, f) Decrescente, g) Crescente, h) Decrescente

4) a) $[-4,-2]$ → Crescente $[-2,1]$ → Decrescente $[1,2]$ Crescente.

b) $[2,-1]$ → Decrescente | $[-1,0]$ → Crescente | $[0,1]$ → Decrescente | $[-1,2]$ → Crescente

c) $(-\infty,0)$ → Crescente | $(0,+\infty)$ → Crescente.

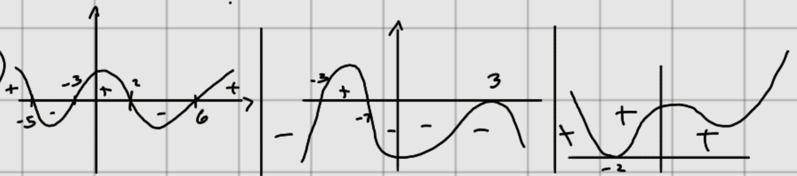
5) a) $y = (m+2)x - 3$
* Crescente $m > -2$
Constante $m = 2$
Decrescente $m < 2$

b) $y = (4-m)x + 2$
* Crescente $m > 4$
Constante $m = 4$
Decrescente $m < 4$

c) $y = 4 - (m+3)x$
* Crescente $m < -3$
Constante $m = -3$
Decrescente $m > -3$

d) $y = m(x-1) + 3 - x \rightarrow y = mx - m + 3 - x \rightarrow x \cdot \underbrace{(m-1)} - m + 3$
* Crescente $m > 1$
Constante $m = 1$
Decrescente $m < -1$

$m = 1$

6) 

7) a) $f(x) = \frac{1}{x-2}$ $D = \{x \in \mathbb{R} \mid x \neq 2\}$

b) $f(x) = \frac{x-1}{x^2-4}$ $D = \{x \in \mathbb{R} \mid x \neq 2 \text{ e } x \neq -2$

c) $f(x) = \frac{1}{\sqrt{x-2}}$ $D = \{x \in \mathbb{R} \mid x > 2\}$

d) $f(x) = \frac{\sqrt{x+2}}{x-2}$ $D = \{x \in \mathbb{R} \mid x > 2\}$

8) $x_v = \frac{-b}{2a}$ $y_v = \frac{-\Delta}{4a}$

a) $(0;4)$ b) $\left(\frac{1}{2}, \frac{1}{36}\right)$ c) $\left(\frac{7}{6}, \frac{-121}{36}\right)$ d) $(1,0)$

9) Encontrando o y_v conseguimos calcular a imagem da função. $y_v = \frac{-\Delta}{4a}$

a) $f(x) = 2x^2 - 8x + 6$ → Para cima $Im = \{y \in \mathbb{R} \mid y \geq -2\}$

$y_v = \frac{-(64-48)}{8} \rightarrow \frac{-16}{8} \rightarrow y_v = -2$

b) $f(x) = -\frac{x^2}{3} + 2x - \frac{5}{3}$

$Im = \{y \in \mathbb{R} \mid y \leq -2\}$

c) $f(x) = x^2 + 4x + 4$ $y_v = \frac{-(16-4 \cdot 4)}{4} \rightarrow 0$

$Im = \{y \in \mathbb{R} \mid y \geq 0\}$

d) $f(x) = -x^2 + 2x - 2$ $y_v = \frac{-(4-8)}{-4} \left.\right\} \frac{-4}{4} \left.\right\} -1$

$Im = \{y \in \mathbb{R} \mid y \leq -1\}$

10) c

11) $P(4,3)$ $-x^2 + H = y \rightarrow -16 + H = 3$ $H = 19$

12) $y = 50 - \frac{x}{2}$ $R = x \cdot y \rightarrow 1250 = x \cdot \left(50 - \frac{x}{2}\right) \rightarrow 1250 = 50x - \frac{x^2}{2} \rightarrow -\frac{x^2}{2} + 50x - 1250 = 0$

$\frac{-50 \pm 0}{-1} \rightarrow 50$

R: Venderam 50 peças.

73) Custa 20 reais o par $x_v = \frac{-b}{2a}$ $y_v = \frac{-\Delta}{4a}$

$(80-x) \to$ Quantidade $F(x) = x.(80-x) - 20.(80-x)$ R: 50 reais gera o lucro máximo.

$F(x) = -x^2+100x-1600 \to y_v = \frac{-3600}{(-4)}\Big\} 900$ $x_v = 50$

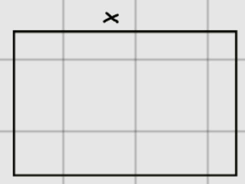
74) $C(x) = 2x^2-7x+8$ $L(x) = R - C \to x^2-x-2x^2+7x-8 \to -x^2+6x-8$

$R(x) = x^2-x$

$x_v = \frac{-b}{2a}$ $\frac{-6}{-2}\Big\} 3$ Deve-se vender 3 unidades.

75) Per $= 20$ $2x+2y=20 \to x+y=10 \to y=(10-x)$

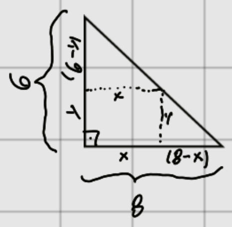
$x.y \to x.(10-x) = f(x) \to F(x) = -x^2+10x$



$y_v = \frac{-100}{-4} = 25$

$x_v = \frac{-10}{-2} \to 5$ Para área máxima $x=5$ e $y=5$

76)



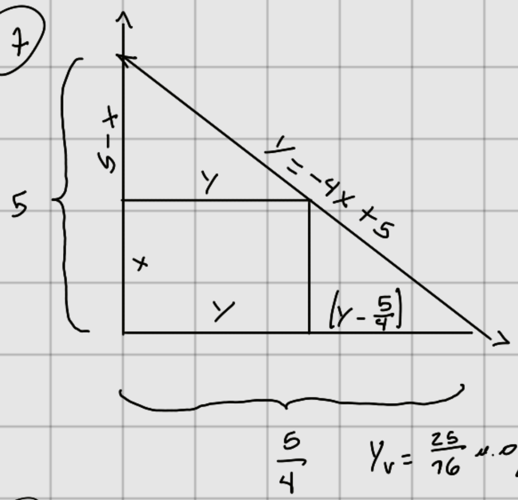
$\frac{6-y}{6} \times \frac{x}{8}$

$6x = 48-8y \to x = -\frac{4}{3}y+8$ $x.y \Rightarrow \left(-\frac{4}{3}y+8\right).y \to \boxed{-\frac{4}{3}y^2+8y = A(y)}$

Como queremos a área máxima. Usaremos o y_v para descobrir o valor que é na imagem.

$y_v\ \frac{-\Delta}{4a} \to \boxed{12\ u.a.}$

77)



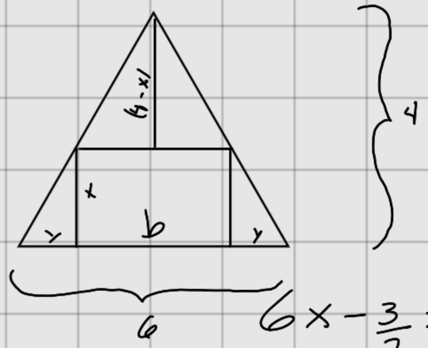
$\frac{y}{\frac{5}{4}} = \frac{5-x}{5}$

$5y = \frac{25}{4} - \frac{5x}{4}\Big\} y = \frac{\cancel{25}-\cancel{5}x}{\cancel{20}}\Big\} y = \frac{5-x}{4}$

$x.\left(\frac{5}{4} - \frac{x}{4}\right) = A(x) \to \frac{5}{4}x - \frac{x^2}{4} = A(x)$

$y_v = \frac{25}{16}$ u.a.

78)



$\frac{6}{4} = \frac{b}{(4-x)}$ $b.x = \left(\frac{12-3x}{2}\right).x \to 6x - \frac{3}{2}x^2$

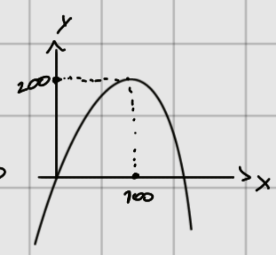
$4b = 24-6x \to b = \frac{3.(4-x)}{2}\Big\} b = \frac{12-3x}{2}$

$6x - \frac{3}{2}x^2 \to y_v = \frac{-36}{-6} \to 6\ u.a.^2$

79)

$x.y =$ Área $x.(400-2x) = A(x) \to A(x) = -2x^2+400$

$2x + y = 400 \to y = 400-2x$ $x_v = \frac{-400}{-4}\Big\} 100$



$\frac{y}{x} = 2$ $y = 200$

(20) $\begin{cases} \frac{1}{x} + \frac{1}{y} = \frac{7}{12} \\ x \cdot y = 12 \end{cases} \rightarrow \frac{x+y}{x \cdot y} = \frac{7}{12} \rightarrow x + y = 7$ $\boxed{x = 7 - y}$

$y.(7-y) = 12 \rightarrow -y^2 + 7y - 12 = 0$

$\frac{-7 \pm 1}{-2} \rightarrow \frac{-8}{-2} \rightarrow 4$

$\rightarrow \frac{-6}{-2} \rightarrow 3$

$S = \{(3,4);(4;3)\}$

(21) $fog \rightarrow f(g(x)) = (2x-3)^2 + 4.(2x-3) - 5 \rightarrow 4x^2 - 12x + 9 + 8x - 12 - 5 \rightarrow 4x^2 - 4x - 8$

$gof \rightarrow 2x^2 + 8x - 10 - 5 \rightarrow 2x^2 + 8x - 15$

$fog(2) = 4x^2 - 4x - 8 \rightarrow 16 - 8 - 3 = 0$

$gof(2) = 9$

Valores de fog que resultam em $y = 16$

$4x^2 - 4x - 8 = 16 \rightarrow 4x^2 - 4x - 24 = 0 \rightarrow 4.(x^2 - x - 6) = 0$

ou $x = 3$

ou $x = -2$

(22) $fog \rightarrow (x-3)^2 + 2 \rightarrow x^2 - 6x + 11$ | $gof \rightarrow x^2 - 1$ | $gog \rightarrow (x-3).x - 3 \rightarrow x^2 - 3x - 3$ | $fof \rightarrow (x^2+2)^2 + 2 \rightarrow x^4 + 4x^2 + 6$

(23) $f(x) = x^3 - 3x^2 + 2x - 1$

Para $f(-x) \Rightarrow f(-x) = (-x)^3 - 3(-x)^2 + 2(-x) - 1 \Rightarrow \boxed{-x^3 - 3x^2 - 2x - 1}$

Para $f(x-1) \Rightarrow f(x-1) = (x-1)^3 - 3.(x-1)^2 - 2.(x-1) - 1 \Rightarrow \boxed{x^3 - x - 1}$

Para $f(\frac{1}{x}) \Rightarrow f(\frac{1}{x}) = (\frac{1}{x})^3 - 3(\frac{1}{x})^2 - 2.(\frac{1}{x}) - 1 \Rightarrow \boxed{\frac{1}{x^3} - \frac{3}{x^2} - \frac{2}{x} - 1}$

(24) $x^2 - 2 = (x+1)^2 - 2 \rightarrow x^2 = x^2 + 2x + 1 \rightarrow x = -\frac{1}{2}$

(25) $fog(x) = 9x^2 - 3x + 1$ $ø = g(x)$

$f(x) = 3x - 5 \rightarrow 3ø - 5 = x^2 - 3 \rightarrow 3ø = x^2 + 2 \rightarrow g(x) = \frac{x^2+2}{3}$

(26) $fog(x) = 9x^2 - 3x + 1 \rightarrow$

$g(x) = 3x - 2$

(27) $f(x) = 2x + 1$ $2ø + 1 = x^2 - 2x + 3 \rightarrow 2ø = x^2 - 2x - 4 \rightarrow ø = \frac{x}{2} - 2x - 4$

$fog(x) = x^2 - 2x + 3$ $g(x) = x^2 - x - 2$

$g(x) = ø$

(28) $f(x) = 2x - 3$ $g(x) = ø$ $2ø - 3 = 2x^2 - 4x + 3 \rightarrow ø = \frac{2x^2 - 4x + 6}{2} \rightarrow g(x) = x^2 - 2x + 3$

$fog = 2x^2 - 4x + 3$

(29) $g(x) = 2x + 3 \rightarrow y = 2x + 3 \rightarrow \boxed{x = \frac{y-3}{2}}$

* Nesse tipo de questão é necessário interpretar o domínio e a imagem para o desenvolvimento da questão.

$fog = \frac{2x+6}{x+1}$

$f(x) = \frac{2.\left(\frac{y-3}{2}\right) + 5}{\frac{y-3}{2} + 1} \Big\} \frac{y - 3 + 5}{\frac{y-3+2}{2}} \Big\} \frac{(y+2)}{y-1} . 2 \Big\} \frac{2y+4}{y-1} \Big\} f(x) = \frac{2.(y+2)}{y-1}$

(30) a) $A(2,3) \rightarrow (3,2)$ b) $B(-2,6) \rightarrow (6,-1)$ c) $(-5,4) \rightarrow (4,-5)$

(31) a) $f(x) = 2x - 5 \rightarrow x = 2y - 5 \rightarrow y = \frac{x-5}{2}$

b) $f(x) = \frac{x+1}{x-4} \Big\}$ $x = \frac{y+1}{y-4} \Big\}$ $x.(y-4) = y + 1 \Big\}$ $\frac{1}{x} = \frac{(y-4)}{(y-1)} \Big\}$ $x = \frac{y-1}{y-4}$

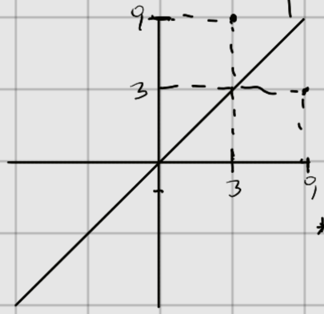
c) $g(x) = x^3 + 2 \rightarrow x = y^3 + 2 \rightarrow y = \sqrt[3]{x-2}$

d) $f(x) = \sqrt[3]{x^2} \rightarrow x = \sqrt[3]{y^2} \Big\}$ $x^3 = y^2 \rightarrow y = \sqrt{x^3}$

e) $f(x) = \sqrt[3]{1-x^3} \rightarrow x = \sqrt[3]{1-y^3} \rightarrow x^3 = 1 - y^3 \rightarrow -y^3 = x^3 - 1 \rightarrow y^3 = -x^3 + 1 \rightarrow y = \sqrt[3]{-x^3+1}$

32

$x = y \rightarrow$ Bissetriz dos quadrantes ímpares



$f(x) = 2x + 3 \qquad f(x)^{-1} = \dfrac{x-3}{3}$

$(3, 9) \qquad (9, 3)$

* Podemos observar que para toda $f(x)$ existe um $f(x)^{-1}$ como mostrado no exemplo numérico